

The Confidence Trap: Financial (Over)confidence and Financial Behaviour Among Italian Adults

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Abstract—Financial literacy is widely recognised as a determinant of individual financial well-being, yet less attention has been paid to how objective knowledge relates to self-perceived competence. Using the Bank of Italy’s IACOFI 2023 survey (N = 4,862), this study examines whether Italian adults with below-average objective financial knowledge who nonetheless rate their own competence as average or higher, a condition termed financial overconfidence, exhibit less prudent financial behaviour or greater exposure to adverse outcomes. This mismatch characterises 15.1 percent of the population, with higher probabilities in the primary model among adults aged 30 to 49 and, separately, among those with upper-secondary education. Among respondents with low objective knowledge, weighted logistic models adjusting for demographic covariates show that higher self-assessed competence is associated with a higher probability of reporting financial incidents (7.9 to 22.0 percent) and a lower probability of careful purchasing (58.3 to 33.5 percent), with weaker evidence in the primary model for more reported long-term goal-setting. All relationships are associational; the cross-sectional design precludes causal interpretation.

1. INTRODUCTION

How much do people really know about money, and how much do they merely think they know? The gap between the two is often where financial mistakes come from. The OECD treats financial literacy as a core determinant of individual economic well-being, defining it as “a combination of financial awareness, knowledge, skills, attitudes and behaviours necessary to make sound financial decisions and ultimately achieve individual financial well-being” [6], and tracks it through a composite indicator built on three pillars: knowledge, behaviour and attitudes, designed for cross-country comparison [7]. In Italy, the Bank of Italy has run this survey since 2017 under the name IACOFI (*indagine sull’Alfabetizzazione e le Competenze Finanziarie*), with the 2023 wave interviewing nearly 5,000 adults aged 18 to 79 [4]. The headline number for 2023 is a modest improvement: an overall score of 10.7 out of 20, up from 10.2 in 2020. This improvement comes from behaviour and attitudes; the knowledge component actually declined slightly, from 3.9 to 3.7.

Keywords: Financial literacy - Financial overconfidence - Self-assessment bias - Survey data - Weighted logistic regression.

AI tools (Claude and ChatGPT) assisted with coding, translation, and LaTeX formatting throughout.

Knowledge scores tell only half the story. The more interesting question is what people *believe* they know, and how that belief lines up with reality. OECD surveys ask respondents to rate their own financial competence against that of their peers, and at the G20 level the answer is strikingly uniform: most people rate themselves as “about average,” a response consistent both with statistical common sense and with a simple reluctance to stand out. Beneath that flat aggregate, though, the literature is far less reassuring. Self-perceived and objective financial knowledge diverge systematically, and in one direction more than the other: people tend to think they know more than they actually do [5]. Psychologists have a name for a version of this pattern outside finance, the Dunning-Kruger effect, where limited competence also limits one’s ability to recognise that competence is limited [3]. Whether that exact mechanism transfers cleanly to money matters is debatable, but a dedicated strand of research has turned the underlying idea into something measurable: “financial overconfidence,” defined as the gap between self-assessed and objectively tested knowledge [1]. The pattern that emerges is almost paradoxical. Greater confidence in one’s own financial knowledge is associated with a lower likelihood of seeking professional financial advice [2]. In other words, feeling knowledgeable may discourage people from looking for help, even when there is still a gap to close.

That paradox is the starting point for this work, and it frames two questions. First, is there an identifiable subgroup of Italian adults with below-average objective financial knowledge and a self-assessment of competence that this knowledge does not support, a genuinely overconfident segment of the population? Second, among adults with low objective financial knowledge, is a higher self-assessment associated with more reported financial incidents or less prudent financial behaviour, once measured knowledge and sociodemographic characteristics are accounted for?

The answer matters for policy, not just for theory, and two institutions in particular have a stake in it: the Bank of Italy, which runs IACOFI and has made financial education part of its remit for over a decade, and Italy’s Committee for Financial Education, the national body charged with coordinating financial education initiatives

across the country. An intervention aimed at raising financial literacy only works if it reaches its audience, and recognising a gap in one's own knowledge can be a reason to take part. Someone who is aware of what they don't know can be a receptive audience for financial education, while someone who overestimates their own competence may not see the need. If a meaningfully overconfident subgroup exists, outreach that relies on self-recognised need risks missing people who would benefit from support.

One caveat runs through everything that follows. IACOFI is a cross-sectional survey, meaning that each respondent is observed at a single point in time, with no way to track how knowledge, self-perception, or behaviour evolved beforehand or afterward. That design rules out causal claims. It cannot establish that overconfidence *leads to* riskier financial choices, nor can it rule out that both are shaped by some third, unobserved factor, such as personality, past financial experience, or the household environment in which one's relationship with money first took shape. Every relationship reported here, adjusted or unadjusted, is therefore a statistical association and should be read as such. Survey weights are applied throughout the analysis.

2. METHODOLOGY AND DESCRIPTIVE ANALYSIS

2.1. Data

The data come from IACOFI 2023 (Surveys on Financial Literacy and Digital Financial Skills in Italy: Adults), run using the standardised OECD/INFE instrument for measuring financial literacy and financial inclusion [7]. The sample counts 4,862 adults, aged 18 to 79, and the questionnaire covers a lot of ground: sociodemographic background, personal financial planning, ownership and choice of financial products, objective financial knowledge, and a battery of financial attitudes and behaviours.

Every respondent carries a sampling weight (*wght*), which corrects for how well the sample represents the Italian adult population. Unless stated otherwise, every descriptive estimate and proportion in this work is weighted; raw, unweighted counts are reported separately wherever sample size itself is the point.

2.2. Variables Used

At the heart of the analysis sit two numbers for each respondent: what they actually know, and what they think they know. The first, objective financial knowledge, comes from seven financial knowledge questions drawn straight from the OECD/INFE toolkit (QK3, QK4, QK5, QK6, QK7_1 through QK7_3). They probe the basics: whether inflation erodes purchasing power, how interest works on an informal loan, the difference between simple

and compound interest, the trade-off between risk and return, and the logic of spreading savings across different products. Answers are simply tallied into a score from 0 to 7,

$$S_i = \sum_{j=1}^7 \mathbb{1}(\text{item}_{ij} \text{ credited}), \quad (1)$$

with anything short of correct, including a “don't know,” counted as a miss. Following the OECD/INFE scoring rule, the compound-interest item receives a point only when both the compound-interest answer and the preceding simple-interest answer are correct [9], on the reasoning that understanding compound interest without simple interest first is not really understanding it at all.

The second number is self-perception, captured by a single question (QK1) that asks respondents to place their own financial knowledge on a five-point scale relative to other people. The original item runs backwards from what feels intuitive, with 1 meaning the highest self-assessed competence and 5 the lowest, so it is flipped before use,

$$A_i = 6 - \text{QK1}_i, \quad (2)$$

giving a scale where higher now means more confident, as most readers would expect. It is a subjective judgement, nothing more, but it is exactly the ingredient that turns a plain knowledge score into a measure of overconfidence: what matters is not the score itself, but how it compares with what the same person believes about themselves.

Behaviour is where the stakes become concrete, and here the analysis draws on four separate outcomes. The first outcome records whether a respondent reported at least one of four financial incidents in the previous two years: an investment scam, disclosure of financial information through a deceptive communication, unauthorised use of card details, or a query about an unrecognised transaction. A second and third capture two habits generally read as prudent, namely thinking carefully about affordability before a purchase and actually setting long-term financial goals rather than just having them in mind. The fourth, shopping around before choosing a financial product, works differently: it only applies to people who made such a choice in the two years before the interview, so it is treated as a secondary outcome with its own, smaller universe rather than folded into the main comparison. Careful purchasing is coded as 1 for “always” or “often” considering affordability and 0 for the remaining substantive responses. Long-term goals are coded as 1 for the two agreement categories and 0 for the other substantive responses. The incident indicator requires valid answers to all four items and equals 1 when at least one incident is reported. Shopping around equals 1 for comparing options within or across providers and 0 for the other substantive choices. Non-substantive responses

are treated as missing.

Alongside all this sit the usual suspects among sociodemographic covariates: age, gender, education, macro-region, household income, how many financial products a respondent already holds, and whether they have internet access. The main models adjust for age, gender, education and macro-region. Additional specifications examine the role of financial product holdings and internet access, while household income is included in a separate sensitivity analysis because of its substantial non-response. These comparisons assess how the main associations change under additional adjustment.

2.3. Strengths and Limitations of the Data

The survey's main strength is the OECD/INFE instrument itself. It is standardised, which makes the financial knowledge measures comparable across countries, and it comes with sampling weights designed to improve the sample's representation of the Italian population aged 18–79. At nearly 4,900 respondents, the sample is also large enough to support solid descriptive estimates for the main demographic subgroups.

The limitations fall into two categories. The first is one already flagged: a cross-sectional design cannot support causal inference. The second is more practical. Some variables carry non-trivial missingness, most notably household income, with an unweighted non-response rate of 31.1 percent, and the shopping-around variable for financial products, which only applies to respondents who actually made a product choice in the past two years, narrowing its effective universe considerably. Both issues called for a tailored approach further down in the modelling strategy.

2.4. Objective and Self-Rated Financial Knowledge

Before any threshold or mismatch definition enters the picture, it is worth looking at each measure on its own terms, because they behave quite differently. Objective knowledge and self-perception are not the same kind of scale, and treating them as interchangeable from the start would obscure more than it reveals.

Objective knowledge varies across the full seven-question scale. The weighted average sits at 3.69 out of 7, with 9.1 percent of respondents scoring zero and only 6.7 percent getting every question right. The distribution peaks at a score of 5 (20.8 percent), which happens to be exactly the OECD/INFE threshold for adequate knowledge; altogether, 39.4 percent of the sample reaches that bar. In other words, most Italians land somewhere in the middle of the scale, and a clear majority fall short of what the OECD/INFE toolkit considers a minimally adequate level of financial knowledge.

Self-rated knowledge tells a different kind of story, and not just because the scale is shorter. Where objective scores fan out across the full range, self-ratings pile up hard in the middle: 45.2 percent of respondents place themselves at exactly "average," with roughly a quarter each below that at "very low" (25.4 percent) and "low" (24.9 percent), and only a thin sliver, 4.4 percent combined, willing to call their own financial knowledge "high" or "very high." Altogether, just under half of respondents (49.6 percent) rate themselves as average or better. A mean is deliberately not used as the primary summary here, since there is no reason to assume the distance between "low" and "average" is the same as between "average" and "high"; the shape of the distribution, not a single average number, is what matters.

Figure 1 puts the two side by side, and the contrast is the point. Objective knowledge looks something close to a bell curve, gently rising and falling across the full range of possible scores. Self-rated knowledge looks nothing like that: it is heavily concentrated at "average" and below, with the confident tail nearly empty. On its own, this contrast does not yet say anything about whether individual respondents are overestimating or underestimating their own competence; that question needs the two measures compared directly, which is what the next section does. But it already suggests something worth keeping in mind going forward: high self-reported confidence is rare in this sample. If a mismatch between knowledge and self-perception exists, it is far more likely to look like quiet, moderate overestimation clustered around "average" than like a loud minority proclaiming expertise it does not have.

2.5. Knowledge–Confidence Profiles and Alternative Mismatch Definitions

With the two distributions established on their own terms, the next question is what happens when they are compared directly, respondent by respondent. This is where the idea of a mismatch between what people know and what they think they know actually gets defined, and it turns out there is more than one reasonable way to do it.

The starting point is a simple grid: whether a respondent's objective score meets the OECD/INFE adequacy target of 5 or above, crossed with whether their self-rating falls below average, exactly average, or high to very high. The middle category, average, is kept separate on purpose rather than folded into either extreme, since the self-rating question asks people to compare themselves with others, and a genuinely neutral answer is not the same thing as modesty or bravado. Figure 2 lays out all six resulting cells. All six percentages refer to the full sample with valid information on both knowledge measures ($n = 4,453$). A third of the sample (33.4 percent, $n =$

1,246) falls below the adequacy target and also rates itself below average, a group with little apparent mismatch at all. A further 22.8 percent of the joint-valid sample fall below target but rate themselves as average, and a further 2.1 percent ($n = 106$) go as far as high or very high. On the other side of the target, 16.9 percent ($n = 736$) meet the adequacy bar but still rate themselves below average, which is its own kind of miscalibration, just running in the opposite direction; 22.4 percent ($n = 1,167$) land on average, and 2.3 percent ($n = 120$) call themselves high or very high.

This same grid can be read through three different lenses, and each one answers a slightly different question. The first, an unrecognised adequacy gap, groups together everyone below the adequacy target who rates themselves average or higher, an absolute standard borrowed directly from the OECD/INFE benchmark. It identifies 24.9 percent of the sample ($n = 1,184$). The second, relative overestimation, asks a narrower question: not whether someone falls short of an external target, but whether they know less than the typical Italian and still rate themselves at least average. Following the approach used by the Bank of Italy in its own financial literacy work [8], this means an objective score of 3 or below, the integer floor of the weighted national mean of 3.69, paired with a self-rating of average or higher. It identifies a smaller, more conservative 15.1 percent of the sample ($n = 712$), and it is this definition that anchors the modelling that follows, precisely because it is grounded in where people actually fall relative to their peers rather than in an externally imposed cutoff. The third, a strict mismatch, tightens the self-rating side instead of the score side: the same below-target respondents, but only those calling themselves high or very high. It leaves just 2.1 percent of the sample ($n = 106$) and is retained as a descriptive sensitivity benchmark.

None of these three should be swapped for one another. The gap between the first two, 24.9 percent against 15.1 percent, is entirely accounted for by a single group: respondents who scored exactly 4, sitting above the national average but still short of the OECD/INFE bar. They alone make up 9.8 of the 24.9 percentage points, 472 of the 1,184 respondents flagged by the broader definition. That is not statistical noise; it marks out a genuine borderline zone between two defensible ways of drawing the same line, and it is exactly why the three definitions are kept visible side by side rather than collapsed into one headline number. Demographic profiling in what follows covers the first two definitions; the third is reported only in aggregate, since 106 cases are too thin to break down safely by age, gender, education, or region.

One caveat is worth stating plainly before moving on. A respondent falling into one of these mismatch cate-

gories is, by construction, a descriptive fact about how their knowledge and self-perception compare, nothing more. It does not yet say anything about whether that mismatch coincides with riskier financial choices or worse outcomes; that connection, if it exists, is what the outcome analysis further ahead is designed to test.

2.6. Demographic Profiles of Knowledge–Confidence Mismatch

Knowing that two overlapping but distinct forms of mismatch exist in the population raises an obvious next question: who, specifically, falls into them? This section breaks both the unrecognised adequacy gap and the relative-overestimation measure down by age, gender, education, and macro-region, always looking at prevalence within each group rather than what share of the mismatched group a given demographic makes up, since the latter is driven as much by how many people are in a group as by anything about the mismatch itself. The strict definition is left out of this breakdown entirely: with only 106 cases nationwide, splitting it further by four demographic dimensions would leave most cells too thin to say anything reliable.

Age tells a fairly consistent story across both definitions. Respondents aged 30 to 49 show the highest prevalence of both the unrecognised adequacy gap (27.2 percent, against an overall 24.9 percent) and relative overestimation (17.5 percent, against an overall 15.1 percent), while the oldest group, 70 to 79, sits noticeably below the national figure on both measures (19.2 percent and 12.4 percent respectively). If anything, this cuts against the easy assumption that overconfidence is a young person's problem: it is the middle of the age distribution, not its youngest end, where the gap between what people know and what they think they know is widest.

Gender shows a small but consistent tilt: men score higher than women on both definitions, though the gap narrows considerably under the stricter, relative-overestimation measure. Under the unrecognised adequacy gap, men sit at 27.6 percent against 22.4 percent for women, a difference of over five points; under relative overestimation, the gap shrinks to 15.9 percent against 14.3 percent, under two points. Education is where the two definitions diverge most sharply, and the divergence is informative rather than just noise. Tertiary-educated respondents show a mismatch prevalence of 27.3 percent under the broad adequacy-gap definition, well above the 19.7 percent seen among those with lower secondary education or less, but slightly below the 28.4 percent seen among those with upper-secondary education, the single highest group on this measure. But once the stricter relative definition is applied, tertiary-educated respondents drop to 13.2 percent, below the national average of 15.1

percent, while upper-secondary respondents remain the highest group at 18.3 percent. In other words, a meaningful share of tertiary-educated respondents flagged by the broad measure are not below-average performers on the objective knowledge test; they are people who fall just short of the OECD/INFE bar for adequate knowledge while still knowing more than the typical Italian, exactly the score-4 boundary group already identified earlier. Macro-region, by contrast, shows only modest variation on either measure, with all five areas sitting within a few points of the national figures.

Figure 3 presents both measures side by side across all four demographic dimensions, with a dashed reference line marking each measure's overall prevalence. None of these comparisons should be read as evidence of a cause. They describe where mismatch concentrates within the population as observed, not why it concentrates there, and a lower prevalence in one group is not the same thing as that group being financially safer.

2.7. Weighted Outcome Distributions and Analytical-Sample Availability

Before any of the four behavioural outcomes can be linked back to the knowledge-confidence mismatch, it is worth pausing on what each one actually looks like on its own, and on how much of the sample is available to describe it. The two questions are connected: an outcome asked of everyone behaves differently, in terms of both its meaning and its missingness, from one asked only of a subset.

A reminder of direction is necessary before looking at the numbers, precisely because it is easy to get backwards. Financial incidents are coded so that 1 means something bad happened; the other three outcomes are coded so that 1 means the respondent behaved prudently. Nothing here shares a common substantive direction just because all four use the same 0/1 coding.

Financial incidents are the least common of the four events by a wide margin: 14.4 percent of respondents (675 out of 4,531 with a valid answer) report having experienced at least one of the four incidents in the battery. Considering affordability before a purchase is a much more widespread habit, reported by 48.5 percent of respondents (2,172 out of 4,554). Setting and actually pursuing long-term financial goals is less common, at 35.7 percent (1,544 out of 4,320), which is unsurprising given that intending to save for the future and consistently acting on that intention are two different things. Shopping around sits differently again, both in its prevalence and in what it can be compared to: among the 1,082 respondents who both made a recent product choice and gave a substantive answer, 49.3 percent report having compared options before deciding, but this figure describes only that self-selected group of recent choosers, not the population

as a whole.

That restriction is worth dwelling on, because it is by far the largest gap in availability among the four outcomes. Financial incidents and the two attitudinal outcomes are all asked of essentially the full sample, with unavailability in the low hundreds, 331 missing for incidents, 308 for affordability, 542 for long-term goals, largely ordinary item non-response. Shopping around is different in kind: the question was only put to the 1,329 respondents who said they had chosen a financial product in the past two years, and of those, 247 gave no substantive answer, leaving 1,082 valid observations out of a starting sample of 4,862. It is a secondary outcome not because it matters less, but because a result based on it can only ever speak to recent product choosers, a group that is not representative of the wider population by definition.

Figure 4 shows all four weighted prevalences together, with the secondary shopping outcome visually set apart from the three drawn from the full sample. Read together with the availability figures just discussed, it sets up the outcome analysis that follows: three outcomes with broad sample support and one useful but narrower one, all still describing the population as observed, with no adjustment yet for age, education, or any of the other covariates that come next.

2.8. Knowledge Calibration and Unadjusted Outcome Profiles

This final descriptive step does two things: it looks at how the two knowledge measures line up across their full range, not just through the lens of a threshold, and it takes a first, still entirely unadjusted look at whether the six knowledge-confidence profiles differ in their behavioural outcomes. These descriptive comparisons establish the patterns that the adjusted models will examine.

Figure 5 lays out the full picture: for every objective score from 0 to 7, it shows how respondents at that score rated their own knowledge. Among respondents who scored zero, most did rate themselves “very low” (62.9 percent), but 15.1 percent still called themselves “about average.” At the other end, among those who got a perfect score, nearly one in five (19.0 percent) still rated themselves “very low,” and only about half (49.9 percent) landed on “average,” suggesting the mismatch runs in both directions even if overestimation, concentrated in the lower rows of the table, is the more common story. A weighted counterpart to Spearman's correlation, which avoids assuming equal spacing between the self-rating categories, puts a single number on the overall consistency between the two measures: 0.25, positive but modest, confirming that objective knowledge and self-assessment move together only loosely across the sample as a whole.

With the calibration picture in view, the natural next

question is whether any of this connects to what people actually do. Figure 6 breaks all four behavioural outcomes down across the same six knowledge-confidence profiles used earlier, still with no adjustment for age, education, or anything else. The clearest pattern shows up in the primary outcome, financial incidents. Among respondents below the adequacy target, the group that rates itself below average reports incidents at 11.4 percent, close to the 14.8 percent reference for this analytical sample. But within that same below-target group, the incident rate jumps to 21.9 percent among those rating themselves as average and rises further, to 26.8 percent, among those rating themselves as high or very high, although that top cell rests on only 97 respondents and is flagged accordingly. The pattern is less uniform where the group meets the adequacy target: incident rates sit at 13.3 percent among those who call themselves below average, dip slightly to 12.7 percent at “average,” and rise again to 20.5 percent among the smaller group rating itself high or very high. Read across both rows, the clearest gradient appears below the adequacy target, where low knowledge paired with a higher self-rating lines up with more reported incidents.

Careful purchasing behaviour tells a related but distinct story. Among respondents below the adequacy target, 53.0 percent of those who rate themselves below average report weighing affordability carefully, compared with only 36.2 percent of those in the same knowledge band who rate themselves average, a reversal in the opposite direction of what the incident numbers showed: here, the respondents who rate their knowledge below average are the more careful ones, not the less careful ones. The remaining cells, along with the long-term-goals and shopping-around outcomes, follow their own patterns of similar magnitude, visible in full in the figure; none of the four outcomes moves in a perfectly uniform way across all six profiles, and some cells, particularly the smallest “high or very high” groups, carry the sparse-cell caveat shown in the figure notes.

These patterns motivate the adjusted analysis: does self-assessment remain associated with financial incidents and purchasing caution after accounting for measured knowledge and demographic differences?

3. ADJUSTED STATISTICAL MODELLING AND RESULTS

3.1. Model Specification and Estimability Audit

Before fitting a single model, it is worth freezing exactly what will be estimated, on what sample, and checking that the resulting design is actually solvable. This audit stage fits nothing; it only locks down specifications and verifies that each one has enough support in the data to

be trusted once it is fitted.

Two families of models are specified. The first profiles those who fall into the knowledge-confidence mismatch, using the Banca-aligned relative-overestimation definition as the primary target, since it is the one grounded in where people actually sit relative to their peers rather than an external benchmark. The broader unrecognised adequacy gap is kept only as a threshold sensitivity under the same core specification, and the strict mismatch is left out of the modelling entirely: with only 106 cases nationwide, a multivariable model built on it would rest on ground too thin to support.

The second family looks at the four behavioural outcomes themselves, and here the specification is built directly around the project’s central hypothesis. Each model includes the objective-knowledge score, centred at the overall survey-weighted mean, the four-level self-rated knowledge variable (very low, low, average, and high or very high, with the top two categories merged because only 12 respondents rated themselves “very high”) and, crucially, their interaction. The interaction allows the relationship between objective knowledge and each outcome to vary across self-rating categories on the logit scale, giving the model more flexibility to examine the combination of knowledge and self-perception at the heart of this project.

Adjustment follows a deliberate hierarchy rather than a single fixed set of covariates. The core specification adjusts for age group, gender, education, and macro-region, the four sociodemographic characteristics available for essentially the whole sample. An exposure-aware extension adds the number of financial products held and internet access, on the reasoning that someone’s actual exposure to financial products and digital channels plausibly shapes both their knowledge and their behaviour. A separate income-adjusted branch adds household income, kept apart from the main specification precisely because of the 31.1 percent non-response on that variable discussed earlier.

Each extension is paired with a core-only anchor, a model using the exact same restricted sample but without the added covariates. The logic is to separate two things that are easy to conflate: comparing the full-core model against its anchor shows what changes when the sample itself shrinks to those with valid exposure or income data, while comparing an anchor against its paired extension shows what changes when a covariate is added on an otherwise identical sample. Conflating the two would risk attributing a shift in the estimates to the wrong cause, sample composition versus actual adjustment, when in fact both are moving at once. Altogether, the frozen registry holds 26 specifications: five for the primary mismatch profiling model, one threshold-sensitivity variant, and

five for each of the four behavioural outcomes.

The estimator itself is a weighted logistic regression, fitted separately for each of the 26 specifications, with survey weights rescaled to a mean of one within each model's own analytical sample. Uncertainty is handled with a respondent-level HC1 sandwich covariance, which does not assume the model is perfectly specified and is more forgiving of the kind of weight-driven heteroskedasticity that survey data typically carries. For respondent i , the robust score contribution is

$$u_i = \tilde{w}_i x_i(y_i - \hat{p}_i), \quad (3)$$

and the sandwich covariance estimate is

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}) = \frac{n}{n-p} B^{-1} \left(\sum_{i=1}^n u_i u_i^T \right) B^{-1}, \quad (4)$$

where

$$B = \sum_{i=1}^n \tilde{w}_i \hat{p}_i(1 - \hat{p}_i) x_i x_i^T. \quad (5)$$

These remain weighted associational models, not causal models and not tools for individual-level prediction; the coefficients themselves are kept mainly for reproducibility, while the results discussed in what follows are expressed as adjusted probabilities rather than raw coefficients.

Before any of the 26 models is actually estimated, each specification's design matrix is checked for full column rank, meaning that no covariate or category is a redundant combination of the others and every parameter is, in principle, estimable. All 26 pass this check. Of the 26, only four carry report priority: the full-core profiling model for the primary mismatch, and the full-core models for financial incidents, careful purchasing, and long-term goals. Shopping around remains secondary throughout, and every anchor or income/exposure-extended model is a technical sensitivity check rather than an additional substantive finding in its own right; fitting 26 models does not mean this project is reporting 26 separate results.

3.2. Weighted Logistic Estimation and Essential Diagnostics

With every specification frozen and confirmed estimable, this stage actually fits the 26 registered models: the weighted logistic regression described above, applied once per specification, with survey weights rescaled to a mean of one within each model's own sample.

Before any result is read, each fit has to clear a short list of essential checks: that the optimiser actually converged, that the information matrix has full rank, that every coefficient and covariance entry is finite, that ro-

bust standard errors come out strictly positive, that fitted probabilities fall inside the valid (0,1) range, that the analytical sample matches what was frozen in the audit stage, and that no warning of quasi-complete separation was raised during estimation. These are not judgement calls about whether a result looks interesting; they are pass-or-fail checks about whether the fit can be trusted at all. All 26 models converge, none raises a separation warning, and all 26 pass every check. Fitted probabilities across the full set of models range from about 0.03 to 0.99.

Two contextual statistics come directly out of this stage, kept deliberately as simple descriptive numbers rather than as the output of yet another model. The first restates the primary mismatch prevalence already established earlier: 15.1 percent of the joint-valid population (712 out of 4,453) falls into the Banca-aligned relative-overconfidence group, and this remains the baseline burden estimate that the adjusted profiling in the next stage works from. The second reframes the same 712 respondents from a different angle, one that speaks more directly to the recognition-gap question at the heart of this project: among the 1,607 respondents whose objective score is 3 or below, meaning those who genuinely know less than the typical Italian, 38.7 percent still rate their own knowledge as average or higher. Put plainly, of the people who most need to recognise a gap in their own financial knowledge, close to two in five do not. This second number is reported only as context, not as a new conditional model in its own right.

3.3. Adjusted Probabilities and Stakeholder-Facing Results

The adjusted results sharpen the central finding: among adults with low objective financial knowledge, a higher self-assessment is associated with more reported financial incidents and less careful purchasing in the primary models.

Who is overconfident, once demographics are accounted for. Figure 7 presents adjusted mismatch probabilities for each demographic dimension separately. In the primary model, adults aged 30–49 have the highest estimated probability among age groups, at 17.3 percent (95% CI: 15.0 to 19.5), compared with 12.0 percent among those aged 18–29 (95% CI: 8.5 to 15.6). Upper-secondary education has the highest estimate among education groups, at 18.1 percent (95% CI: 16.5 to 19.7), compared with 12.6 percent for tertiary education (95% CI: 10.3 to 14.9). Gender differences are smaller: 15.9 percent for men and 14.4 percent for women. Macro-region shows relatively little variation. These results place age and education at the centre of the demographic description; their stability across alternative specifications is examined below.

Does the mismatch line up with behaviour, once knowledge itself is held fixed? To answer this question, we compare adjusted outcome probabilities across self-ratings among adults with low objective financial knowledge. The outcome models are first estimated on their full available samples. We then use these models to calculate predictions for respondents scoring 3 or below: 1,427 for financial incidents, 1,488 for careful purchasing, and 1,383 for long-term goals. Within each target sample, self-rating is set in turn to each category while objective knowledge and demographic characteristics remain at each respondent's observed values. Predictions are averaged using survey weights, allowing the self-rating categories to be compared over the same target population within each outcome. Figure 8 brings these comparisons together: does a more reassuring self-assessment go together with safer financial behaviour?

For financial incidents, the pattern is unambiguous in direction. The adjusted probabilities rise with self-assessment: 7.9 percent at “very low”, 13.7 percent at “low”, and 22.0 percent at “average”. The pre-specified Average-minus-Low contrast is 8.2 percentage points (95% CI: 2.7 to 13.7). This is a substantial difference: within the low-knowledge target population, an “average” self-assessment is associated with a higher probability of reporting a financial incident, even after accounting for measured knowledge and the core demographic characteristics.

Careful purchasing behaviour moves in exactly the opposite direction, which is reassuring in its own way, because it suggests the incident finding is not just noise running in whichever direction happens to look interesting. Among low-scoring respondents, the adjusted probability of considering affordability carefully is 58.3 percent at “very low” (CI: 53.8–62.8), falling to 43.2 percent at “low” (CI: 37.6–48.8) and 33.5 percent at “average” (CI: 29.4–37.6). The average-minus-low contrast here is –9.7 percentage points (CI: –16.6 to –2.8), again an interval that does not cross zero, but this time entirely below it. The respondents who correctly recognise how little they know about money are, on this evidence, the more careful shoppers, not the less careful ones; it is the ones who paper over a real gap in knowledge with an “average” self-rating who behave less cautiously.

Long-term goals tell a less conclusive story. The adjusted probability rises from 21.8 percent at “very low” (CI: 17.5–26.2) to 41.2 percent at “average” (CI: 36.9–45.4), and the average-minus-low contrast is a positive 5.6 percentage points, but its interval, –1.3 to 12.5, straddles zero. Here, more confidence is associated with more reported goal-setting, a prudent behaviour, but the pre-specified Average-minus-Low contrast remains inconclusive in the primary model. The high-or-very-high category on all

three outcomes rests on comparatively few respondents (55 to 57 in the low-score subsample) and is flagged accordingly in the figure; its estimates are shown for completeness but should be read as considerably less precise than the very-low-to-average range that anchors the main contrasts.

Shopping around, the secondary outcome, is reported separately and outside these two figures, restricted as always to the smaller universe of recent product choosers; its estimates carry wider intervals and a documented support warning, and are not treated as part of the project's central finding.

Taken together, these adjusted results are the closest this project comes to answering its own title. Among people who genuinely know less than the typical Italian about money, those who do not recognise that gap, rating themselves as average or better anyway, report more financial incidents and behave less carefully before purchases, even after age, gender, education, and macro-region are held constant. The evidence for more reported long-term goal-setting is weaker in the primary model and points to a different aspect of financial behaviour. None of this is causal: the cross-sectional design cannot rule out that some third factor, such as general risk tolerance or personality, drives both the mismatch and the behaviour. But the incident and purchasing patterns persist after demographic adjustment, supporting the project's central hypothesis.

3.4. Robustness Checks

The three pre-specified contrasts and the adjusted demographic profile are checked one last time before drawing any conclusions, this time against changes in sample and adjustment rather than against zero. No model is refitted here; the same 26 specifications from the previous stage are simply compared with one another. For each of the three primary outcomes, the full-core estimate is set against a paired anchor restricted to respondents with valid exposure data (financial products held, internet access) and a second anchor restricted to those with valid income, and each anchor is then compared with its own extended version once the corresponding covariate is actually added. The same exercise is repeated for the fourteen adjusted demographic probabilities behind the mismatch profile.

The direction of all three outcome contrasts holds up across every specification, although their magnitude and precision vary. Careful purchasing still points towards less caution at higher self-ratings, although the contrast weakens in the income-complete sample and its interval then includes zero. Financial incidents provide the most consistent evidence: the positive contrast remains separated from zero in every specification. The long-term-goals

contrast remains positive in every specification too, but its interval excludes zero in both specifications restricted to respondents with income data. Much of the increase is already present in the restricted-sample anchor, making sample composition important to its interpretation. The demographic profile of the mismatch shows more selective stability: upper-secondary education remains the highest-probability education category, while adults aged 30–49 lead in the primary model and those aged 18–29 lead after income adjustment.

The robustness checks therefore refine the hierarchy of evidence: the incident contrast is the most consistent across specifications, purchasing caution remains a supporting result, and long-term goals depend more on which sample is analysed. Full detail on all fifteen outcome-contrast comparisons and the paired demographic estimates is kept in the project's technical outputs rather than reproduced here, in line with keeping this report proportionate to what a stakeholder actually needs to act on.

3.5. Discussion of Results

Having checked which patterns survive changes in sample and adjustment, it is worth restating the story this project set out to test in plain terms before turning to what it can and cannot show. Among Italian adults with below-average objective financial knowledge, financial incidents rise steadily as self-rated knowledge climbs, while careful purchasing behaviour runs in the mirror-image direction. Long-term goal-setting adds a different pattern: more confidence, more reported goal pursuit, but the Average-minus-Low contrast remains inconclusive in the primary model, making this a complementary finding.

This answers, directly, the two questions this project opened with. To the first, whether an identifiable subgroup exists with below-average objective knowledge and a self-assessment this knowledge does not support: yes, and it is not a marginal phenomenon, comprising 15.1 percent of the adult population under the primary definition, or 712 of the 4,453 respondents with valid data on both measures. To the second, whether this condition is associated with less prudent behaviour or greater exposure to adverse outcomes once knowledge itself and sociodemographic characteristics are accounted for: yes for financial incidents and for careful purchasing in the primary models, both with confidence intervals that exclude zero. Long-term goal-setting shows a positive but statistically weaker association in the primary model.

Not all three findings deserve equal weight, and the robustness checks are what make that hierarchy explicit rather than a matter of impression. Financial incidents provide the most consistent evidence across specifications: their contrast stays clearly separated from zero throughout, although it carries a local-support caveat, since part

of the pattern rests on comparatively thin cells. Careful purchasing keeps its negative sign, but its interval includes zero in the income-complete specifications; the main contrast has no sparse-cell warning, while the sensitivity contrasts do. Long-term goals never flip sign either, but their interval excludes zero in the income-complete specifications, with local support warnings remaining. They are best read as a complementary finding whose strength depends on the sample.

This connects directly back to the recognition-gap figure established earlier: among the 1,607 respondents with an objective score of 3 or below, 38.7 percent still rate their own knowledge as average or higher. The adjusted results in this section connect that recognition gap to behaviour by comparing self-ratings among adults with low objective knowledge. Higher self-ratings do not line up with uniformly worse outcomes, but they do accompany more reported financial incidents and, in the primary analysis, less careful purchasing.

The demographic profile adds a second, more practical layer to the same story. In the primary model, adults aged 30 to 49 lead among age groups, and upper-secondary education leads among education categories, challenging a simple deficit-based view centred only on the youngest or least educated. The educational pattern holds up under the robustness checks, while the age ranking changes after income adjustment. This matters for the two institutions named at the outset of this report. An intervention aimed at raising financial literacy has to reach its audience, and the literature reviewed in the introduction already suggested why relying on self-recognised need may be a problem: people with greater confidence in their own knowledge tend to seek out financial advice less often [2]. Taken together with the demographic and outcome findings, this suggests that outreach relying on self-recognised need is likely to systematically miss part of its intended audience, and that combining a brief objective knowledge check with a self-assessment or decision-support prompt, rather than waiting for people to identify their own gap, may reach this group more effectively; any such approach would itself need piloting rather than being adopted on the strength of an associational study alone.

None of this should be overstated. The cross-sectional design established in the introduction still applies with full force: these are adjusted associations, not effects of confidence on behaviour, and the same unobserved factors that could jointly drive both, a person's general risk tolerance, prior financial shocks, or personality traits that shape self-assessment on any topic, remain just as unaccounted for here as they were before adjustment. Adjusting for age, gender, education, and macro-region leaves the incident and purchasing associations in place, but it does not rule out confounding by unmeasured fac-

tors. The weaker result for long-term goals in the primary model is a reminder that even under the same adjustment strategy, the three outcomes do not provide equally clear evidence.

4. CONCLUSIONS

This study set out to answer two questions using the Bank of Italy's IACOFI 2023 survey. First, does a subgroup of Italian adults exist with below-average objective financial knowledge and a self-assessment this knowledge does not support? Yes: 15.1 percent of the adult population, 712 of 4,453 respondents with valid data, falls into this category, and among those who score lowest on objective knowledge, 38.7 percent still do not recognise the gap. Second, is this mismatch associated with less prudent behaviour or greater exposure to adverse outcomes, once knowledge itself and sociodemographic characteristics are accounted for? Also yes, most clearly for financial incidents, whose contrast holds up across every sample and adjustment tested. Everyday purchasing prudence provides a second signal, although it weakens in the income-complete sample. Long-term goal-setting points towards more reported prudent behaviour, with weaker evidence in the primary model and clearer evidence in the income-complete specifications.

The demographic profile points consistently to upper-secondary education, while the age ranking is less stable. More broadly, the findings give the two institutions a reason to reach people who may not recognise their own educational needs. This suggests that financial education initiatives, including those coordinated by the Bank of Italy and Italy's Committee for Financial Education, may do better to pair a brief objective knowledge check with a self-assessment prompt rather than relying on self-recognised need. Respondents could then be directed towards practical content on recognising financial risks and checking affordability before spending. Demographic patterns can inform outreach, while the joint assessment provides a direct starting point for discussing individual educational needs. This approach gives the analysis a concrete use: making gaps between demonstrated knowledge and self-perception visible when financial education is offered. Its value should be evaluated through a pilot comparing the proposed approach with a standard invitation, measuring engagement, completion of the educational activity, and subsequent financial decision-making.

These findings are associational, not causal. The cross-sectional design cannot rule out that some unobserved factor, such as risk tolerance or personality, drives both the mismatch and the behaviour, and the results should be read accordingly. A natural next step would be to test whether this pattern replicates in a future wave of IACOFI or in panel data tracking the same individuals over time.

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Objective and self-rated financial knowledge

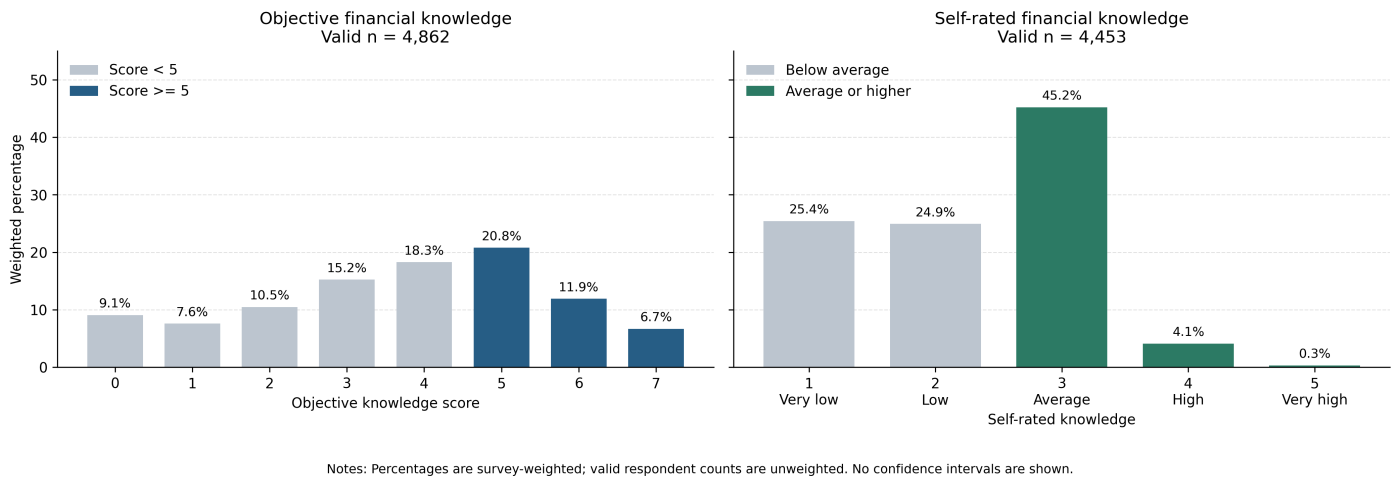


Figure 1. Objective financial knowledge score (0–7) and self-rated financial knowledge (1–5), survey-weighted.

Note: Percentages are survey-weighted; respondent counts (n = 4,862 for objective knowledge, n = 4,453 for self-rated knowledge) are unweighted. Bars highlighted in colour mark, respectively, scores meeting the OECD/INFE adequacy target (objective knowledge, score ≥ 5) and self-ratings of average or higher (self-rated knowledge, rating ≥ 3). No confidence intervals are shown.

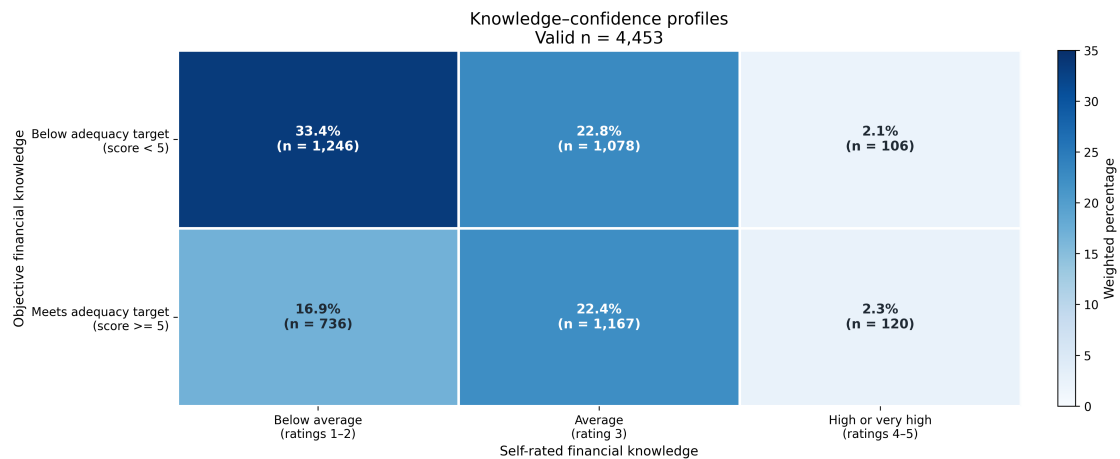
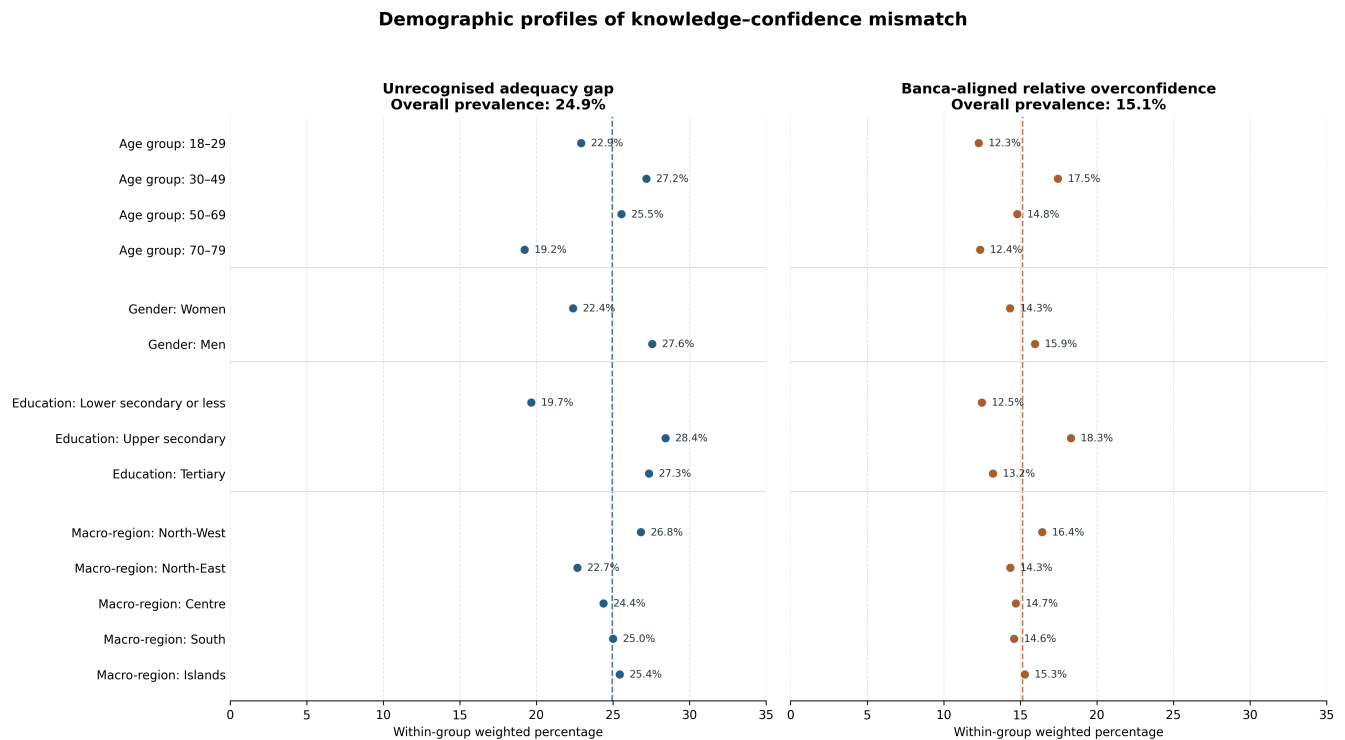


Figure 2. Joint distribution of objective knowledge adequacy and self-rated financial knowledge (survey-weighted percentages; joint-valid n = 4,453).

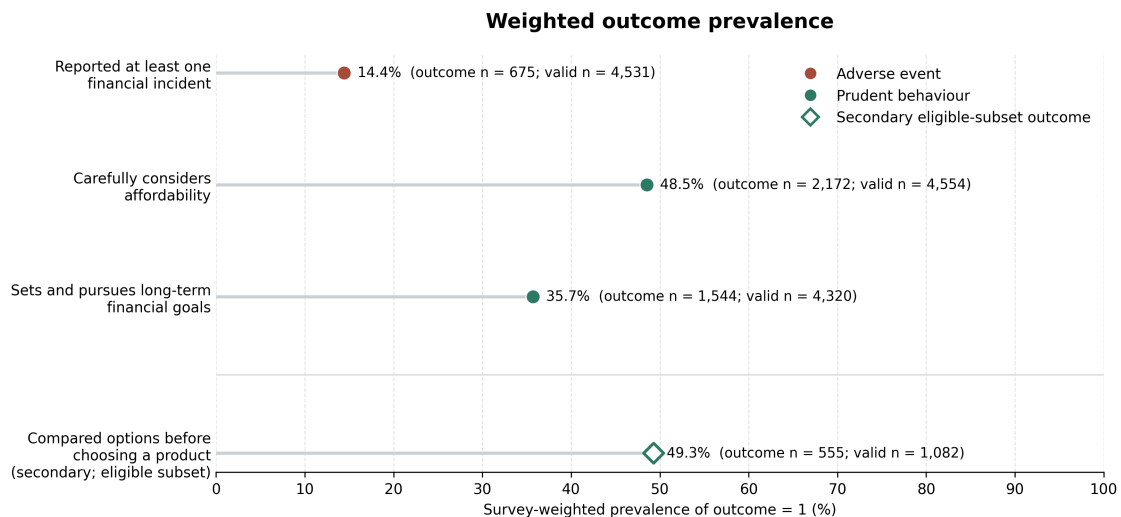
Note: Percentages are survey-weighted; respondent counts are unweighted. The neutral “Average” self-rating category is retained separately rather than merged with either extreme, and no single cell is labelled as overconfidence.



Notes: Percentages are survey-weighted within each demographic category; counts in the saved table are unweighted. Dashed lines show each measure's overall prevalence. Descriptive estimates; no confidence intervals or significance tests are shown.

Figure 3. Weighted prevalence of the unrecognised adequacy gap and Banca-aligned relative overconfidence, by age group, gender, education, and macro-region.

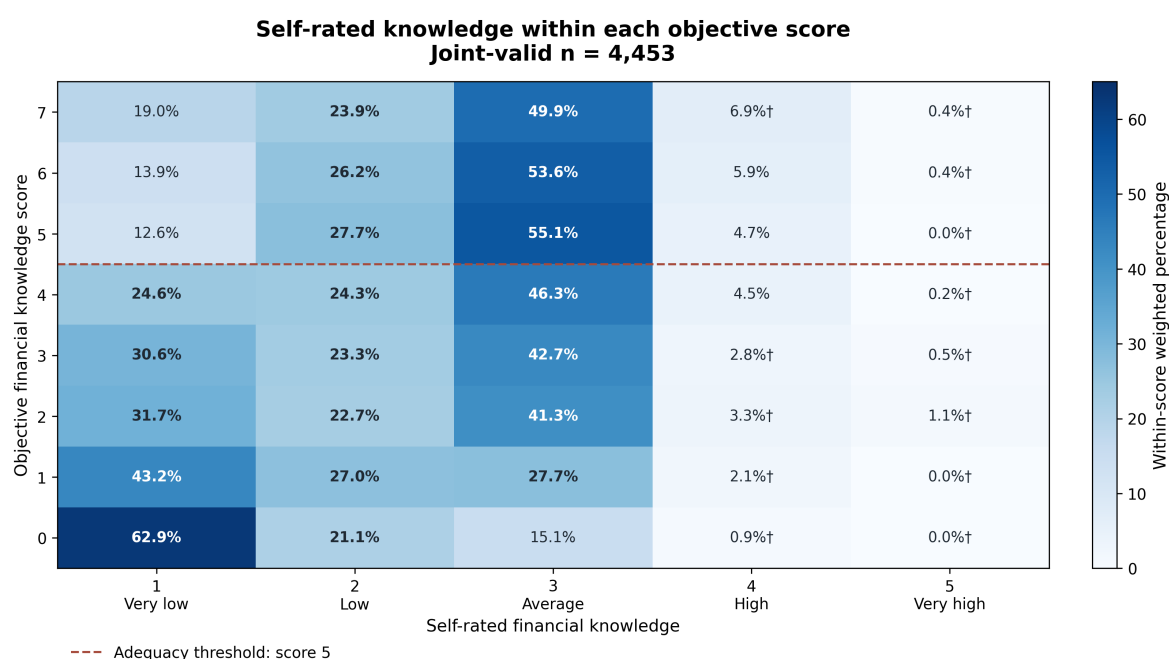
Note: Percentages are survey-weighted; respondent counts are unweighted. Dashed lines mark each measure's overall prevalence (24.9 percent for the unrecognised adequacy gap; 15.1 percent for Banca-aligned relative overconfidence). Descriptive estimates only; no confidence intervals or significance tests are shown.



Notes: Percentages are survey-weighted within each outcome-specific valid base; counts are unweighted. Outcome = 1 denotes an adverse event for financial incidents and a prudent behaviour for the other outcomes. Shopping around is restricted to its observed eligible subset. No confidence intervals or tests are shown.

Figure 4. Weighted prevalence of the four behavioural outcomes.

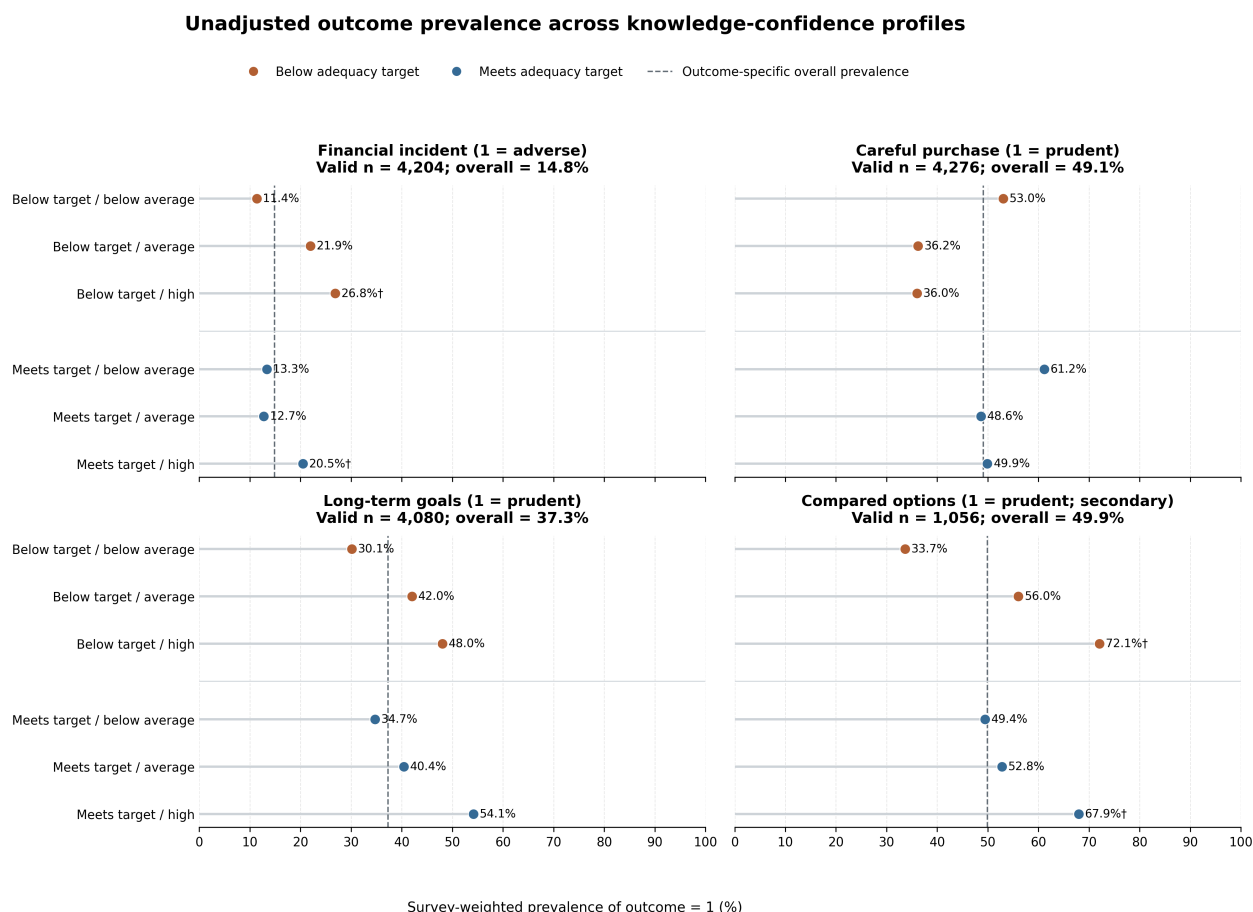
Note: Percentages are survey-weighted; respondent counts are unweighted. Outcome = 1 denotes an adverse event for financial incidents and a prudent behaviour for the other three outcomes. Shopping around is restricted to its observed eligible subset (recent product choosers) and is not comparable to the other three, which are asked of the full sample. No confidence intervals are shown.



Notes: Each row sums to 100% apart from rounding. Percentages are survey-weighted; † indicates an unweighted cell n below 30. Cell counts are retained in the saved table.

Figure 5. Self-rated financial knowledge within each level of objective financial knowledge (survey-weighted percentages by row; joint-valid $n = 4,453$).

Note: Each row sums to 100 percent. The dashed line marks the OECD/INFE adequacy threshold (objective score of 5 or above). A dagger (†) flags cells resting on fewer than 30 unweighted respondents. Percentages are survey-weighted; underlying counts are unweighted.



Notes: Percentages are survey-weighted within each profile and outcome-specific valid base. Results are unadjusted and are not causal comparisons. † flags a profile denominator below 50 or a binary outcome count below 30; no cell is excluded. Shopping around is restricted to its eligible subset. Counts are retained in the saved table.

Figure 6. Unadjusted weighted prevalence of the four behavioural outcomes across the six knowledge-confidence profiles.

Note: Percentages are survey-weighted; respondent counts are unweighted. Outcome = 1 denotes an adverse event for financial incidents and a prudent behaviour for the other three outcomes. Shopping around is restricted to its eligible subset. Results are unadjusted and are not causal comparisons; a dagger flags a profile denominator below 50 or a binary outcome count below 30. No confidence intervals or significance tests are shown.

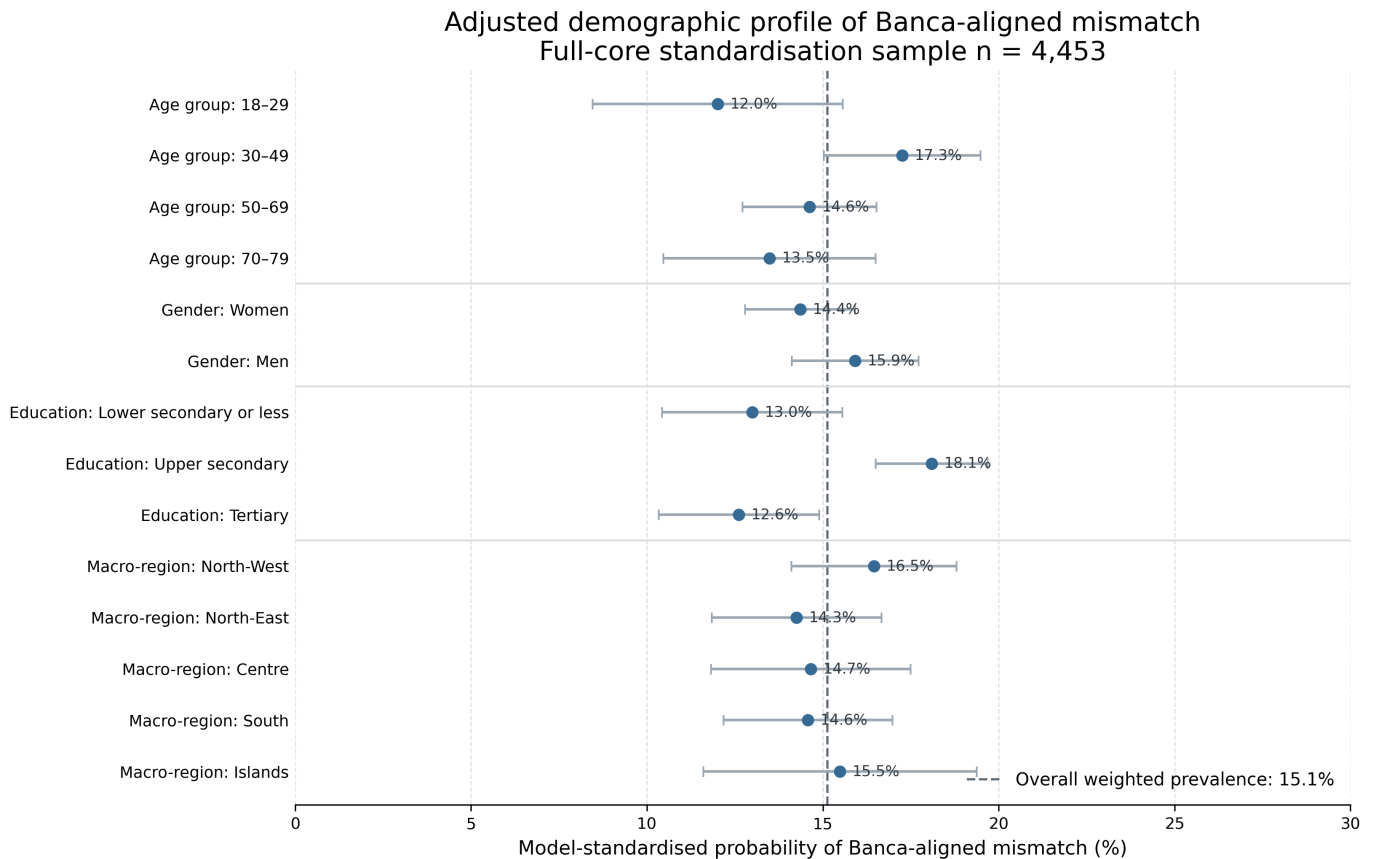


Figure 7. Model-standardised probability of Banca-aligned relative overconfidence, by age, gender, education, and macro-region, with 95% delta-method confidence intervals.

Note: Probabilities are standardised to a common covariate distribution using the full-core profiling model; the dashed line marks the overall survey-weighted mismatch prevalence of 15.1 percent. Estimates describe adjusted associations, not causal effects.

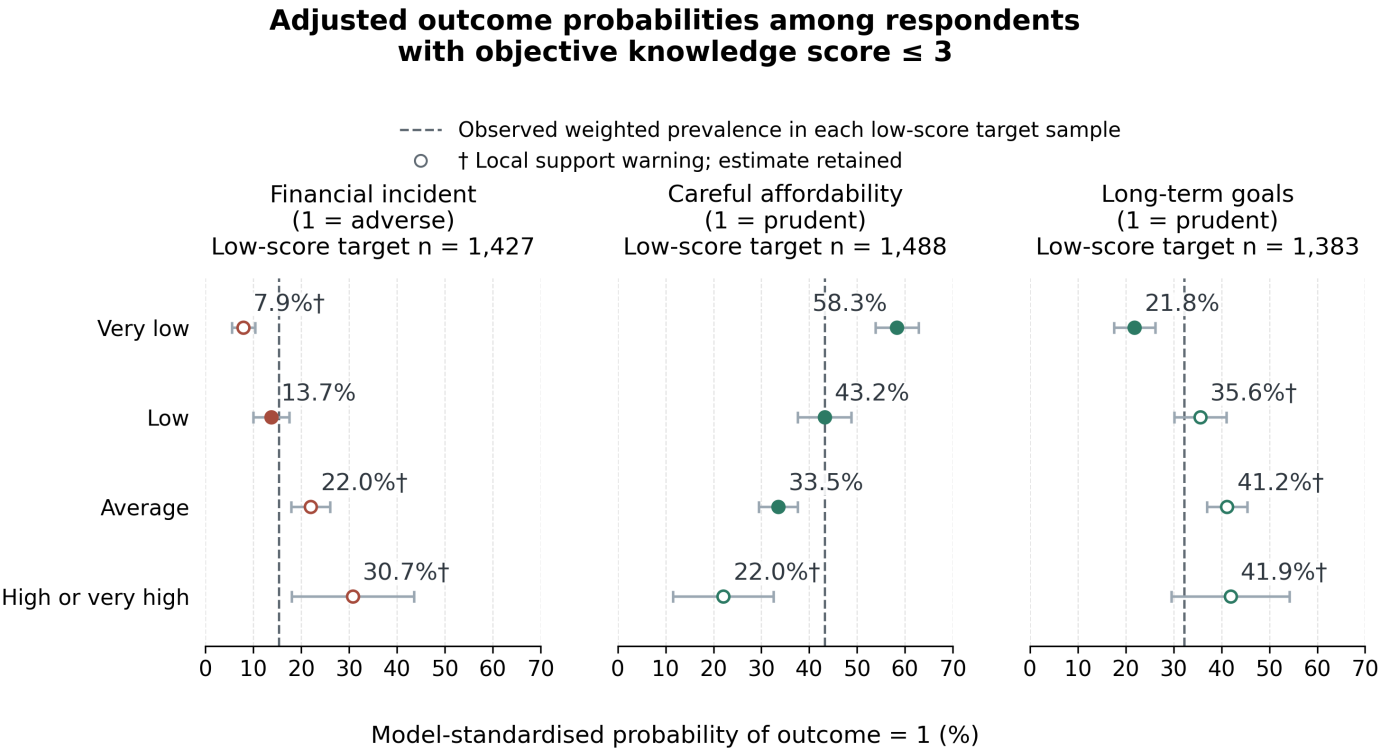


Figure 8. Adjusted probability of each primary outcome among respondents with objective knowledge score ≤ 3 , by self-rated knowledge, with 95% delta-method confidence intervals.

Note: Estimates hold objective score and all core covariates at each respondent’s observed values. A dagger flags estimates with limited local data support. Results describe adjusted associations among people who already score low on objective knowledge; they are not causal effects of confidence itself.